

Birla Institute of Technology and Science, Pilani
Digital Signal Processing (EEE/ECE F434)

Mid-Semester Examination, First Semester 2024 – 25

Time: 90 minutes

03/10/2024

Maximum Marks: 60

Answer all questions in sequence and attempt all parts of a question together

1. Justify the following with reasoning, no marks if no justification.

[Reasoning/Justification: brevity is the key, it saves paper and time]

- (a) You must be aware that an asteroid from the Arjuna asteroid belt will be earth's second 'mini-moon' for the next few weeks. As a DSP engineer you have been given the task to transmit its captured image. At the receiver, the received signal is $x[n]$; for $x(n)$ which component has more important perceptual information: the DFT magnitude or phase. (3)

- (b) For an input $x[n]$, suppose we decimate by a factor of 2 and get the output as $y[m]$. Next, we create a shifted-by-one-sample version of $x[n]$ and call it $x_{shift}[n]$, and then decimate $x_{shift}[n]$ by a factor of 2 to obtain $y_{shift}[m]$.

Claim: "The decimation process is time invariant if $y_{shift}[m]$ is equal to a time-shifted version of $y[m]$." Prove the correctness of this claim. (4)

- (c) A CDMA-based wireless communication system has 10 Mega chips per second (Mcps) code rate and a 4.8 kbps information data rate. If the spreading code generation rate is increased to 50 Mcps, how much improvement in the processing gain of CDMA system will be achieved. Is there any advantage in increasing the spreading code generation rate with a 4.8 kbps information data rate? (3)

- (d) Suggest a procedure to perform linear convolution of three sequences of lengths N_1 , N_2 , and N_3 by DFT. (3)

- (e) "In OFDM-based 5G wireless systems, by appending a CP to the channel input, the linear convolution associated with the channel impulse response becomes a circular convolution." Prove the correctness of this statement mathematically. (4)

2. Consider $x[n] = [1, 0, -1, \underline{2}, 3]$, where the underline indicates the origin. The DTFT is denoted as $X(w) = X_R(w) + jX_I(w)$, where $X_R(w)$ and $X_I(w)$ are real and imaginary parts of $X(w)$ in rectangular coordinates. Find the signal $y[n]$ with DTFT given as $Y(w) = j \int_{2\pi} X_R(\Phi)X_I(\Phi - w) d\Phi$. Hint: use DTFT properties. (7)

For the following parts, marking scheme is not specified. You are supposed to decipher it based on the bulleted questions given at the end, and that will give you (12) marks.

3. The pole-zero diagram in Figure 1 corresponds to $X[z]$, the Z-transform of a causal sequence $x[n]$. Give the pole-zero diagram of $Y[z]$, where $y[n] = x[-n + 3]$, also specify the ROC for $Y[z]$. (?)

4. Figure 2 illustrates a six point DT sequence $x[n]$, assume that $x[n]$ is zero outside the shown interval. The value of $x[4]$ is not known and is given as b . Let $X(e^{jw})$ be the DTFT of $x[n]$ and $X_1[k]$ be samples of $X(e^{jw})$ at $w_k = 2\pi k/4$, $0 \leq k \leq 3$. The sequence $x_1[n]$ obtained by taking the IDFT of $X_1[k]$ is shown in Figure 3. Can you

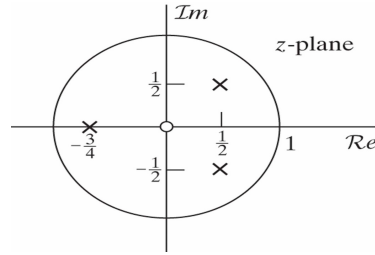


Figure 1: Pole zero plot.

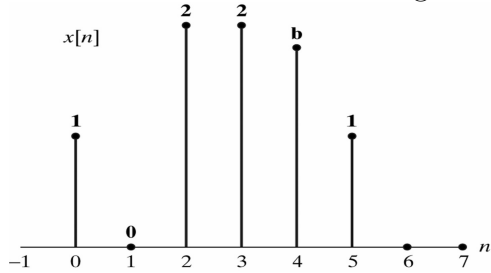


Figure 2: DT sequence $x[n]$.

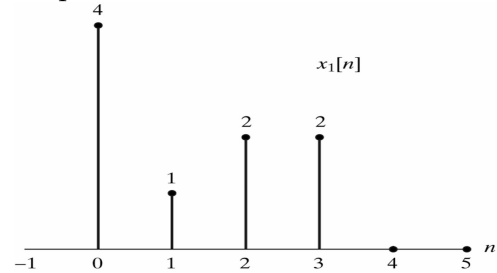


Figure 3: DT sequence $x_1[n]$.

determine b uniquely, if so, give the value of b . Does this sampling cause time domain aliasing ?

(??)

- In the lecture, we had discussed that the basic butterfly flow-graph of the DIT radix-2 FFT algorithm which is given in Figure 4. Develop the butterfly diagram for $N = 8$ when the input and output are in normal order. Comment on the advantage/disadvantage of the developed butterfly computation flow-graph.

(???)

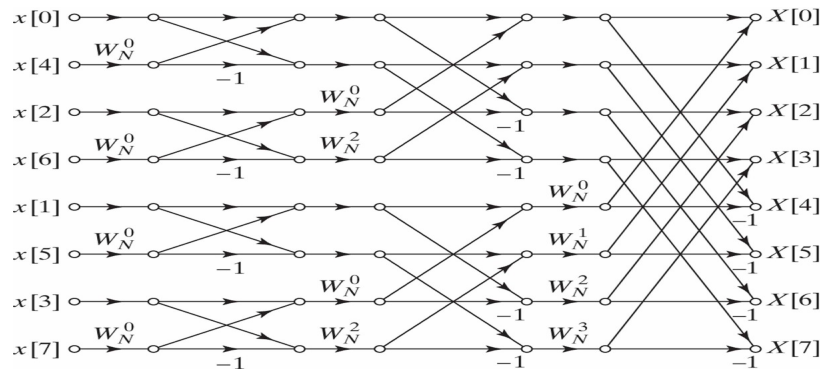


Figure 4: DIT butterfly discussed in the lecture, in case you do not remember.

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- Given $X[k] = [6, 8 + 4j, -2, 8 - 4j]$, find $x[n]$ using matrix method. Then, $\sum x[n] = [?]$.
 - A 4-point signal $[1, 2, 3, 4]$ has DFT $X[k]$, find the time domain signal corresponding to $[X[k] + X^*[k]]$ using DFT properties. The sum of first and second sample of the obtained time domain signal gives [??].
 - Compute mod-3 circular convolution of $h = [1, 2, -1, 1]$ and $x = [1, 1, 2, 1, 2, 2, 1, 1]$, then the first sample of the result obtained gives [???].
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